

Experiment 3 Convolution (Convolution Sum) - Part I

I. Experiment Objectives

1. Understand the concepts of sampling and time delay;
2. Examine how convolution represents the system response as a superposition of impulse responses.

II. Pre-Lab Preparation

1. Review the operation of pulse generators and sequence generators;
2. Review the definition of convolution and the convolution sum;
3. Answer the following questions in the lab report:
 - ① Write down the formula and computational steps for convolution integral;
 - ② Express the convolution of a signal $f(t)$ with an impulse function.

III. Experimental Principles

1. Convolution

The convolution integral method allows the zero-state response of a linear time-invariant (LTI) system to be computed using its impulse response. The zero-state response $r(t)$ is given by the convolution of the input signal $e(t)$ and the system's impulse response $h(t)$, expressed as:

$$r(t) = e(t) * h(t) \quad (3.1)$$

The computation of the convolution integral involves four steps: Folding, Shifting, Multiplication, Integration. The key to evaluating the integral is determining the correct integration limits.

Given two functions $f_1(t)$ and $f_2(t)$, the integral $f(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$ is called the convolution integral of $f_1(t)$ and $f_2(t)$, denoted as

$$f(x) = f_1(t) * f_2(t)$$

2. Algebraic Properties of Convolution

- (1) Commutative Property: $f_2(t) * f_1(t) = f_1(t) * f_2(t)$
- (2) Distributive Property: $f_1(t) * [f_2(t) + f_3(t)] = f_1(t) * f_2(t) + f_1(t) * f_3(t)$

The distributive property applies to parallel systems: the overall impulse response of a parallel connection equals the sum of the individual subsystem impulse responses.

- (3) Associative Property: $[f_1(t) * f_2(t)] * f_3(t) = f_1(t) * [f_2(t) * f_3(t)]$.

The associative property applies to cascaded systems: the overall impulse response of a series connection equals the convolution of the individual subsystem impulse responses.

IV. Experimental Procedure

1. Impulse Response Measurement (Lab 5)

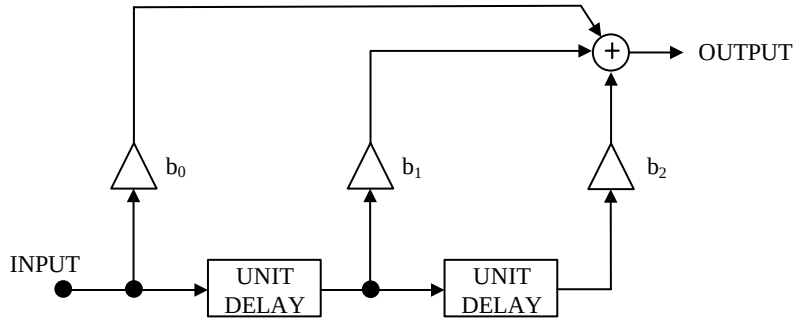


Figure 3.1 Unit impulse delay.

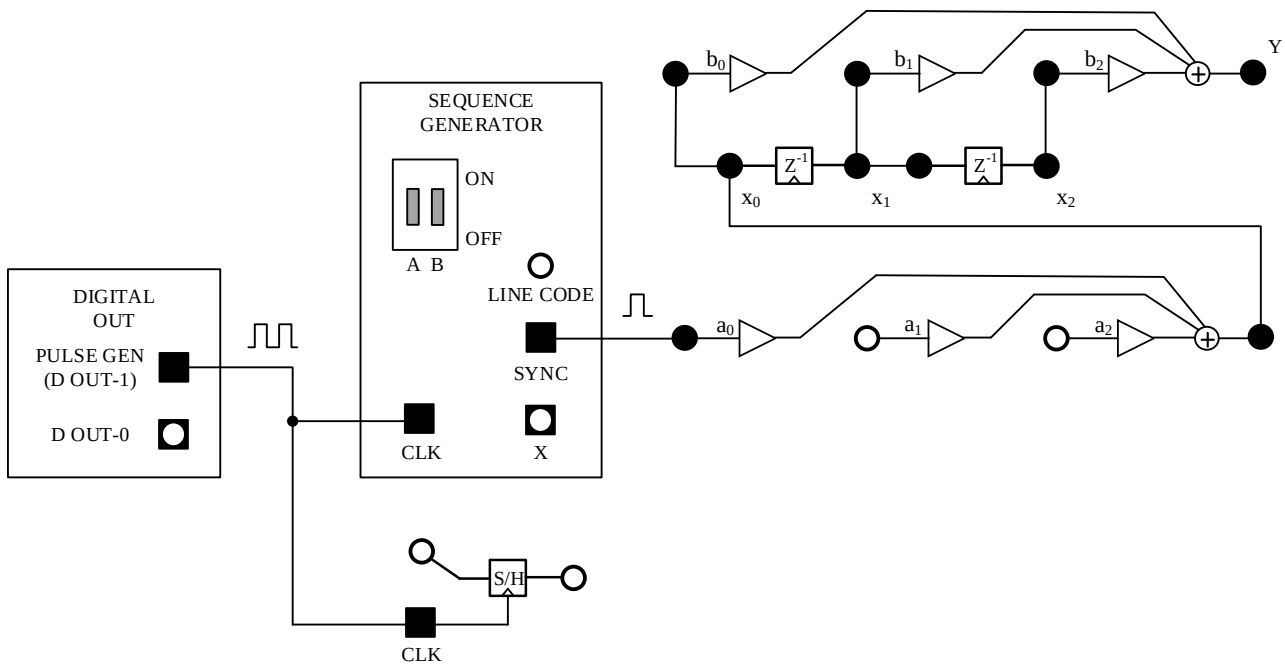


Figure 3.2. Connection diagram of the unit impulse response.

Launch the SIGEx software, connect the circuit as shown in Figure 3.2. and set the following parameters:

① Pulse generator: Frequency: 1 kHz, Duty cycle: 0.5 (50%).

② Sequence generator: Set both DIP switches UP.

③ Oscilloscope: Set the time base according to the signal requirements. Use both oscilloscope channels to observe the system input and output simultaneously. Set the trigger source to the system input (CH0), with rising-edge triggering and a trigger level of 1 V.

The SYNC output of the sequence generator (SEQUENCE GENERATOR SYNC) has an amplitude of 5 V. Use gain amplifier a_0 to reduce this amplitude. Adjust the gain of a_0 while observing the signal on the oscilloscope so that the pulse amplitude (i.e., the output of adder “a”) is exactly 1 V. Record the value of a_0 .

Set $b_0 = 0.3$, $b_1 = 0.5$, $b_2 = -0.2$. Display the output of the adder “b” on the oscilloscope. Measure and record the amplitude of each pulse in the output sequence ($h(0)$, $h(1)$, $h(2)$).

The experiment shows that a single input pulse produces delayed and scaled samples in the delay line. These samples are summed in the adder. The resulting output is the system's response to a single pulse. This response is defined as the unit impulse response $h(n)$, which is the system output when the input pulse has unit amplitude.

2. Input Pulse Pairs

(1) Two Consecutive Pulses

Set the sequence generator DIP switches to UP: DOWN to select a two-pulse sequence. Use the same gain settings as in Step 1, adjust the oscilloscope time base and trigger level as needed, observe the output signal and measure and record the amplitudes of the pulses labeled $h(0)$, $h(1)$, $h(2)$, and $h(3)$.

V. Important Notes

1. General safety and setup procedures are the same as in Experiment 2.
2. When verifying additivity, carefully identify the index of each measured pulse. Plan the measurement before recording data.